

ORCHESTRATION

Orchestration in the Monkey Head Project coordinates Huey's modular components, ensuring that workloads are distributed efficiently, resources are used optimally, and recovery from faults is automated.

Core Tools & Frameworks:

- Docker – Containerizes individual services for isolation, portability, and ease of deployment.
- Kubernetes – Automates deployment, scaling, and management of containerized applications.
- CI/CD Pipelines – Integrate testing and deployment to maintain a rapid development cycle.

Operational Advantages:

- Scalability – Rapidly adjusts resources to meet demand.
- Fault Tolerance – Automatically redeploys failed containers or nodes.
- Modularity – Allows independent updates of services without impacting others.

Governance Integration:

- Orchestration events are recorded in the Unified Audit Log.
- Governance approval is required for scaling beyond predefined thresholds.

Testing Objectives:

- Measure container deployment speed under simulated demand spikes.
- Evaluate system stability during orchestrator upgrades.
- Validate governance alerts for unauthorized configuration changes.